BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

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ABSTRACT

Accurate prediction of drug–target interactions (DTIs) can support the prioritisation of candidate pairs, yet many deep-learning models provide little visibility into the features used for prediction. We present BioInteract, a DTI framework combining a chemistry-aware molecular graph encoder, an ESM-2/physicochemical protein residue encoder with a configured domain-label channel, and bidirectional cross-attention. The model produces atom–residue attention matrices and atom-level Grad-CAM scores as model-native attribution outputs. A full-precision reconstruction from the released random-split checkpoint and inputs achieved an AUROC of 0.904 on the Davis kinase inhibitor benchmark; the corresponding entity-held-out reconstructions are reported with their complete provenance. We additionally audit chemical and sequence similarity across these partitions and evaluate exact-sequence-grouped cold-start protocols. On a strict exact-entity-novel BindingDB cohort, the frozen Davis checkpoint achieved an AUROC of 0.560 and an AUPRC of 0.340, indicating limited external transfer. The attribution analyses generate hypotheses about model-relevant atoms and residues, but do not constitute predicted three-dimensional contacts or structural mechanisms. The released input package contains no curated per-protein domain annotations, so the domain-label channel uses its unknown label and is not treated as domain evidence. BioInteract therefore provides a transparent starting point for prioritisation and for experimentally testable hypotheses, subject to the dataset and structural-validation limitations stated below.

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Keywords

Drug–target interaction; interpretable deep learning; cross-attention; graph neural network; protein language model; model attribution

1 Introduction

The emergence of "AI for Science" has increased expectations that computational models should not merely maximise predictive accuracy, but should also expose the evidence and uncertainty needed for scientific hypothesis generation.[1] This expectation is especially important in drug discovery, where identifying drug–target interactions (DTIs) requires reliable prioritisation together with inspectable outputs that can guide, but not replace, mechanistic follow-up.

Experimental profiling of binding affinities through surface plasmon resonance, isothermal titration calorimetry, or kinase activity assays is laborious and prohibitively expensive. The Davis kinase inhibitor dataset[2] alone required systematic measurement of 72 inhibitors against 442 kinases—a campaign spanning only a fraction of the druggable proteome. Computational DTI prediction can ease this bottleneck by prioritising the most promising candidates and thereby reducing cost and attrition in development pipelines.[3,4] Yet the field has largely remained trapped in a *predictive-accuracy race*: successive deep learning architectures—DeepDTA,[5] GraphDTA,[6] MolTrans,[7] DrugBAN[8]—have incrementally improved benchmark metrics while leaving an important modelling question only partly addressed: *which input features drive a given prediction?*

This question lies at the heart of the broader "AI for Science" agenda. A prediction with no interpretable output is, at best, a hypothesis generator. Moving from *prediction* to hypothesis generation therefore benefits from models that expose internal attribution patterns—for example, attention distributed across drug atoms and protein residues and gradient-based atom scores—rather than returning only a scalar probability. These outputs can prioritise follow-up experiments, but they are not direct measurements of physical contacts or pocket geometry.

Classical DTI methods illuminate the fundamental tension between interpretability and accuracy. Structure-based approaches (molecular docking, molecular dynamics) provide structural inference anchored in three-dimensional protein geometry,[9] yet are constrained by the availability of high-quality crystal structures and the steep computational cost of free-energy calculations. Ligand-based methods (QSAR, molecular fingerprints)[10] circumvent structural requirements at the expense of mechanistic transparency. Neither paradigm fully satisfies the twin demands of modern drug discovery.

Deep learning has transformed predictive capabilities in this domain. Pretrained protein language models such as ESM-2[11] encode evolutionary and structural information at residue resolution, while graph neural networks faithfully capture molecular topology. Concurrently, chemical language models have demonstrated that pretrained molecular representations support accurate and interpretable prediction of reactivity and toxicity,[12] and chemistry-enhanced transformer architectures such as Radical-Net recover atom-level reaction mechanisms.[13] Together, these advances underscore the importance of domain-specific pretraining and chemistry-aware inductive biases in molecular AI;[14] their integration into interpretable DTI frameworks, however, has remained limited.

In this study, "biological prior knowledge" refers specifically to the sequence information encoded by frozen ESM-2 pretraining together with explicit per-residue physicochemical descriptors. It does not denote curated Pfam, InterPro, binding-site, or other protein-domain annotations.

The "AI for Science" paradigm benefits from attribution mechanisms that are part of the prediction pathway rather than appended only after training. Attention visualisation and SHapley Additive exPlanations (SHAP) can nevertheless have faithfulness limitations.[15] A central principle is therefore that, when attention is an integral component of the prediction pathway, the resulting attributions can be examined consistently with model output but still require external validation before mechanistic interpretation.

Motivated by these observations, we present **BioInteract**, a framework that brings model-native attribution outputs to binary high-affinity DTI classification. BioInteract rests on three architectural components:

1. **Chemistry-aware molecular graph encoding.** Drug molecules are represented as attributed graphs encoding functional-group-relevant properties (H-bond donor/acceptor status, hydrophobicity, partial charges) that are available to the learned molecular representation.

2. **Protein residue encoding with a disclosed domain-label limit.** Protein representations fuse ESM-2 residue embeddings with per-residue physicochemical properties and a configured domain-label channel. No curated per-protein domain-annotation files are present in the released Davis input package; the channel therefore receives the unknown label at every position and is not interpreted as Pfam/InterPro information.

3. **Bidirectional cross-attention attribution module.** A multi-head cross-attention mechanism computes a matrix $\mathbf{M} \in \mathbb{R}^{n_{\text{atoms}} \times n_{\text{residues}}}$ of learned attention weights between drug-atom and protein-residue representations. The matrix is an integral component of the forward pass; it is used here as a model-native attribution output, not as a predicted atom-level structural contact map.

We evaluate BioInteract on the Davis kinase inhibitor dataset[2] under a random pair split and two entity-held-out protocols: Drug-ID-held-out and Target-ID-held-out. We report chemical and sequence-similarity diagnostics for these partitions, because identifier-level exclusion alone does not guarantee scaffold- or sequence-level novelty. We also add exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations. The attribution analyses—cross-attention maps, Grad-CAM scores, attention-concentration summaries, and case studies—are interpreted as model behaviour rather than as independent evidence of molecular recognition mechanisms.

2 Related Work

2.1 Deep Learning for DTI Prediction

Deep learning approaches to DTI prediction have evolved through several generations.[16] *Sequence-based* models encode both drug SMILES and protein amino acid sequences as one-dimensional strings. DeepDTA[5] employs parallel one-dimensional convolutional neural networks (CNNs) for drug and protein encoding; WideDTA[17] extends this framework with additional molecular descriptors. Although computationally efficient, these models discard molecular topology and are limited in their capacity to model long-range intramolecular interactions.

Graph-based models overcome this limitation by representing drugs as attributed molecular graphs. GraphDTA[6] benchmarks a graph convolutional network (GCN), graph attention network (GAT), graph isomorphism network (GIN), and hybrid GAT–GCN encoder while retaining a CNN for proteins. MolTrans[7] introduces mutual interaction learning between molecular substructure representations and protein subsequences.

Attention-based models include architectures that learn local or cross-modal representations for DTI/DTA prediction. TransformerCPI[18] applies transformer attention to drug–protein interaction patterns. DrugBAN[8] introduces a dual bilinear attention network for fine-grained local interaction modelling. AttentionDTA[19] employs multi-head self-attention for both drug and protein representations. In most prior work, however, attention operates as a computational device rather than as an interpretable output channel.

2.2 Pretrained Protein Language Models

ESM-2,[11] trained on millions of protein sequences via masked language modelling, produces residue-level embeddings that encode evolutionary conservation, secondary structure propensity, and contact information. These pretrained representations substantially outperform hand-crafted features on downstream tasks.[20] In DTI prediction, pretrained representations have already been combined with graph drug encoders and interaction modules. For example, GEFA uses pretrained protein representations in an early-fusion graph architecture,[21] PGraphDTA replaces a CNN target encoder with protein language-model features,[22] and HCAF-DTA combines protein representations with bidirectional cross-attention.[23] BioInteract therefore does not claim to be the first application of a protein language model, a graph encoder, or cross-attention to DTI/DTA prediction. Its specific contribution is the submitted integration of a chemistry-aware molecular graph encoder, residue-level ESM-2/physicochemical features with a configured domain-label channel, and inspectable model-native attribution outputs for the Davis binary-classification setting. These outputs are not treated as biologically grounded structural evidence without independent validation.

Recent DTA studies also illustrate the use of residue-level protein graphs, domain-specific representations, or biological-context features alongside evaluation on more than one benchmark, including 3DProtDTA,[24] the domain-specific framework of Liu *et al.*,[25] and InceptionDTA.[26] We cite these studies to contextualise the requested external evaluation, rather than treating their reported results as directly comparable baselines under the present Davis binary classification protocol.

2.3 Interpretability in DTI Models

Existing interpretability approaches include post-hoc methods (attention-weight visualisation, gradient-based attribution, and SHAP values[27]) and model-native attribution mechanisms. Attention weights do not necessarily equal feature importance.[15] BioInteract uses a cross-attention matrix that contributes to prediction and can be inspected as a model-native attribution. It does not by itself establish atom-level contacts, binding modes, or physical interaction types. In contrast, models such as MONN use explicit non-covalent interaction supervision derived from structural complexes;[28] no comparable supervision is used here.

2.4 Cold-Start Evaluation

Random-split evaluation permits information leakage through shared drugs or targets between training and test sets, yielding inflated estimates.[29] Cold-start protocols address this by ensuring that test identifiers are absent from training data. We report the original Drug-ID-held-out and Target-ID-held-out protocols alongside the random split, and separately use exact-sequence grouping for stricter within-Davis stress tests.

3 Methods

3.1 Problem Formulation

Given a drug molecule d represented by its SMILES string and a target protein t represented by its amino acid sequence, we formulate DTI prediction as binary classification: predict whether d binds t with high affinity. Following the established protocol for the Davis dataset,[2] continuous K_d values are binarised at 30 nM ($pK_d > 7.52$), classifying pairs with $K_d < 30\,\text{nM}$ as positive interactions.

3.2 Architecture Overview

BioInteract comprises five sequential modules: (1) a graph isomorphism network with edge features (GINE)-based drug encoder producing per-atom representations, (2) a target encoder combining ESM-2 embeddings with physicochemical features and a configured domain-label channel, (3) a bidirectional cross-attention module computing attention matrices between ligand-atom and protein-residue representations, (4) gated attention pooling to aggregate variable-length representations, and (5) a multilayer perceptron (MLP) prediction head. The full model contains 2,442,083 trainable parameters.

3.3 Drug Molecular Graph Encoder

3.3.1 Chemistry-Aware Atom Featurisation

Each atom is represented by a 52-dimensional feature vector encoding:

- **Topological features** (38 dimensions): atom type (one-hot, 20 types), hybridisation state (6), formal charge (6), number of attached hydrogens (6).
- **Structural features** (9 dimensions): degree, aromaticity, ring membership, and ring size membership (3–8).
- **Functional features** (5 dimensions): hydrogen-bond donor and acceptor status, Crippen logP contribution, molar refractivity, and Gasteiger partial charge.

Bond features (16 dimensions) encode bond type, conjugation, ring membership, and stereochemistry.

3.3.2 GINE Message Passing

We employ a 3-layer GINE architecture:[30]

$$\widetilde{\mathbf{e}}_{vu} = \mathbf{W}_e \mathbf{e}_{vu} + \mathbf{b}_e \in \mathbb{R}^{256}, \qquad \mathbf{e}_{vu} \in \mathbb{R}^{16}, \tag{1}$$

$$\mathbf{h}_v^{(\ell+1)} = \mathrm{MLP}^{(\ell)}\left(\mathbf{h}_v^{(\ell)} + \sum_{u \in \mathcal{N}(v)} \mathrm{ReLU}\left(\mathbf{h}_u^{(\ell)} + \widetilde{\mathbf{e}}_{vu}\right)\right) \tag{2}$$

where $\mathbf{h}_v^{(\ell)}$ is the representation of atom v at layer ℓ, $\mathbf{e}_{vu}$ denotes the raw 16-dimensional edge feature vector, and $\widetilde{\mathbf{e}}_{vu}$ is its learned 16-to-256 projection before message passing. The released `GINEConv` instantiation uses the default fixed $\varepsilon = 0$. Each layer incorporates batch normalisation, ReLU activation, dropout ($p = 0.2$), and a residual connection; the hidden dimension is $D = 256$. Global graph pooling is *not* applied at this stage, preserving individual atom representations for the subsequent cross-attention step.

The archived checkpoints use the stated molecular featurisation; public checkpoint reconstruction applies no stochastic graph augmentation.

3.4 Protein Target Encoder

3.4.1 ESM-2 Residue Embeddings

Residue-level representations are extracted from ESM-2 (150 M parameters, `esm2_t30_150M_UR50D`),[11] yielding 640-dimensional embeddings for each amino acid. These embeddings encode evolutionary conservation, secondary structure propensity, and long-range contact information learned from 65 million protein sequences. ESM-2 parameters are frozen throughout training.

3.4.2 Physicochemical Property Encoding

Each residue is augmented with a 4-dimensional physicochemical vector:

$$\mathbf{p}_i = \left[\text{hydrophobicity}_i,\ \text{mol. weight}_i,\ \text{charge}_i,\ \text{polarity}_i\right] \tag{3}$$

Hydrophobicity follows the Kyte–Doolittle scale.[31] These values provide residue-level physicochemical descriptors; by themselves, they do not identify pocket membership, solvent exposure, or interaction partners.

3.4.3 Configured Domain-Label Channel and Availability

The implementation contains a learnable domain-label embedding layer:

$$\mathbf{d}_i = \mathrm{DomainEmbed}(\text{domain\_type}_i) \in \mathbb{R}^{32} \tag{4}$$

The label builder assigns the unknown `NONE` type when a per-protein annotation file is unavailable. The released Davis package contains no such annotation files, so all archived input positions use this unknown label. We therefore do not represent this channel as curated Pfam/InterPro context, as a binding-site label, or as evidence for a domain-specific conclusion.

3.4.4 Feature Fusion

The three feature sources are concatenated and projected through a shared linear transformation:

$$\mathbf{r}_i = \mathrm{Proj}([\mathbf{e}_i \,\|\, \mathbf{p}_i \,\|\, \mathbf{d}_i]) \in \mathbb{R}^D \tag{5}$$

where $\mathbf{e}_i \in \mathbb{R}^{640}$ is the ESM-2 embedding and Proj is a two-layer MLP with layer normalisation and Gaussian error linear unit (GELU) activation. In the released configuration, $\mathbf{d}_i$ is the shared unknown domain-label embedding rather than a curated per-residue annotation. The projected dimension $D = 256$ matches the drug encoder output.

3.5 Cross-Attention Interaction Module

The cross-attention module produces the model-native attribution used in this study. It computes bidirectional attention between ligand-atom and protein-residue representations, producing an explicit atom–residue attention matrix.

3.5.1 Drug-to-Protein Attention

$$\mathrm{Attn}(\mathbf{Q}_d, \mathbf{K}_p, \mathbf{V}_p) = \mathrm{softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_p^\top}{\sqrt{d_k}}\right)\mathbf{V}_p \tag{6}$$

where $\mathbf{Q}_d = \mathbf{H}_d \mathbf{W}_Q^d$, $\mathbf{K}_p = \mathbf{R}\mathbf{W}_K^p$, $\mathbf{V}_p = \mathbf{R}\mathbf{W}_V^p$, with $H = 8$ heads and head dimension $d_k = D/H = 32$.

3.5.2 Protein-to-Drug Attention

Symmetrically, protein residues attend over drug atoms:

$$\mathrm{Attn}(\mathbf{Q}_p, \mathbf{K}_d, \mathbf{V}_d) = \mathrm{softmax}\left(\frac{\mathbf{Q}_p \mathbf{K}_d^\top}{\sqrt{d_k}}\right)\mathbf{V}_d \tag{7}$$

3.5.3 Attribution Map Extraction

The atom–residue attribution map is obtained by averaging drug-to-protein attention weights across all heads:

$$\mathbf{M} = \frac{1}{H}\sum_{h=1}^{H} \mathrm{Attn}_h(\mathbf{Q}_d, \mathbf{K}_p) \ \in \ \mathbb{R}^{n \times L} \tag{8}$$

Because $\mathbf{M}$ is computed within the forward pass rather than post-hoc, it is a model-native attribution matrix. It is not interpreted as a physical contact map or as an assignment of a specific interaction type to an atom pair.

3.6 Gated Pooling and Prediction

Variable-length representations are aggregated via gated attention pooling:

$$\mathbf{z} = \sum_i \alpha_i \cdot g(\mathbf{x}_i), \qquad \alpha_i = \frac{\exp(f(\mathbf{x}_i))}{\sum_j \exp(f(\mathbf{x}_j))} \tag{9}$$

where $f\colon \mathbb{R}^D \to \mathbb{R}$ and $g\colon \mathbb{R}^D \to \mathbb{R}^D$ are learned two-layer MLPs with Tanh activation. The pooled drug and protein representations are concatenated and passed through a two-hidden-layer MLP ($256 \to 128$; ReLU; dropout $p = 0.3$) followed by sigmoid activation to produce the binding probability.

3.7 Training Provenance and Evaluation Thresholds

The archived entity-split checkpoints preserve model architecture but do not store a uniform optimizer, scheduler, warmup, or label-smoothing record. The Random and Drug-ID-held-out logs explicitly record weighted binary cross-entropy, label smoothing of 0.05, AdamW, and a final-run header, whereas the checkpoint-matched Target-ID-held-out trace has the older `run_split.py` header and records only loss and validation AUROC. Its checkpoint does not retain optimizer or loss state. We therefore do not retrospectively assert a single training recipe across all three historical checkpoints (Supporting Information, Training-provenance table).

No historical entity-split model is retrained for the current-release metrics. The canonical reconstruction instantiates each of those three released checkpoints from its embedded model configuration, performs full-precision inference, selects an F1 threshold once on the corresponding validation predictions, and applies that frozen threshold to test predictions. This separation prevents test-set threshold selection.

In contrast, the two exact-sequence-grouped strict protocols were trained from initialisation on their respective new training partitions. They use seed 42, the same Davis preprocessing and architecture, weighted binary cross-entropy with 0.05 label smoothing, AdamW (learning rate 10^{-4}; weight decay 10^{-5}), batch size 64, gradient-norm clipping at 1.0, a five-epoch linear warm-up followed by cosine decay, and a maximum of 100 epochs. The checkpoint with the highest validation AUROC was selected with patience 15, after which an F1 threshold was selected on that protocol's validation predictions and frozen for its test evaluation. Complete strict-split training provenance, checkpoint paths, SHA-256 hashes, selected epochs, and thresholds are provided in the Supporting Information Training-provenance table.

3.8 Interpretability Analysis

3.8.1 Residue-Level Attention Profiling

For a given drug–target pair, per-residue attention scores are obtained by summing the interaction map over all drug atoms:

$$s_j = \sum_{i=1}^{n} M_{ij}, \quad j = 1, \ldots, L \tag{10}$$

These values are model-attribution scores and are not assigned a biological hotspot label.

3.8.2 Grad-CAM Functional-Group Attribution

Grad-CAM[32] is adapted to the graph-neural-network (GNN)setting:

$$\mathrm{CAM}_v = \mathrm{ReLU}\left(\sum_k w_k \cdot A_v^k\right), \quad w_k = \frac{1}{|\mathcal{V}|} \sum_v \frac{\partial z}{\partial A_v^k} \tag{11}$$

where A_v^k is channel k of the third (final) GINE message-passing layer and z is the pre-sigmoid positive-class logit. Gradients are therefore taken with respect to the logit, not the post-sigmoid probability.

Functional groups are detected with predefined SMARTS patterns (aromatic ring, hydroxyl, amino, amide, carboxyl, sulfonyl, halogen, ether, heterocycle N, and carbonyl). For each group, all RDKit substructure matches are merged into a deduplicated atom set and the group score is the mean of the normalised atom-level scores over that set. An atom can match more than one group pattern. These scores indicate sensitivity of the prediction to the learned representation and do not identify chemical bonds or contacts in a three-dimensional complex.

3.8.3 Attention Sparsity

For each drug–target pair, residue scores are normalised by that pair's maximum:

$$\tilde{s}_j = \frac{s_j}{\max_k s_k}. \tag{12}$$

Pair-level sparsity is then

$$\mathrm{Sparsity} = \frac{|\{j : \tilde{s}_j < 0.1\}|}{L}. \tag{13}$$

The reported global value uses all 1,506 positive Davis records, rather than a single test partition, evaluated with the archived `checkpoints/best.pt` checkpoint. Each pair is normalised separately; the 1,301,380 valid-residue scores are then concatenated before the global distribution and fraction are computed. Consequently, all original train/validation/test memberships are represented and longer valid target sequences contribute more residue-level observations.

4 Experimental Setup

4.1 Dataset

The Davis kinase inhibitor dataset[2] comprises 68 drugs (kinase inhibitors) and 442 targets (protein kinases) forming 30,056 drug–target pairs. Binding affinities (K_d) are binarised at 30 nM, yielding approximately 5 % positive interactions.

4.2 Pre-specified External BindingDB Evaluation

To test exact-entity transfer without changing the submitted Davis training protocol, we curated the official BindingDB curated-articles snapshot `BindingDB_BindingDB_Articles_202607`[33]. We retained only finite, exact numerical K_d measurements in nM for valid, single-chain proteins of at most 1,200 residues and RDKit-valid ligands. Inequality-qualified values were excluded. Before replicate reconciliation, any external pair was excluded when its canonical isomeric SMILES occurred anywhere in Davis or when its full normalised amino-acid sequence occurred anywhere in Davis. The resulting cohort is therefore double-novel by exact ligand and target-sequence identity, although it is not a claim of scaffold or remote-homology novelty. Measurements were then grouped by canonical ligand–sequence pair. A pair with replicate measurements on opposite sides of the 30 nM boundary was removed (a pair-level conflict); for every remaining pair, the median exact K_d was retained and binarised as $K_d < 30$ nM. The complete row-to-pair accounting is reported in Table S13.

The released `best_random` checkpoint was evaluated without retraining, fine-tuning, calibration, or threshold selection on BindingDB. We used the existing random-validation threshold of 0.5959881544 and the same 640-dimensional ESM-2 representation used for Davis. Only newly extracted external protein embeddings that passed dimension and sequence-length checks were supplied to the frozen model. AUROC and AUPRC confidence intervals use 600 pair-stratified bootstrap replicates with seed 42. Curation and frozen-inference provenance, including source and prediction hashes, are reported in Table S13.

4.3 Data Splitting Protocols and Similarity Audit

- **Random split** (70/10/20): drug–target pairs are assigned randomly; shared entities may appear across splits.
- **Drug-ID-held-out split**: drugs are partitioned such that all interactions of a given compound identifier appear in exactly one split. This prevents drug-identifier overlap but does not impose a scaffold-similarity threshold.
- **Target-ID-held-out split**: targets are partitioned analogously; test target identifiers are absent from training. This initial entity-held-out protocol does not impose a sequence-similarity threshold.

For the Drug-ID-held-out audit, we computed non-chiral radius-2, 1024-bit Morgan fingerprints with RDKit and reported the maximum Tanimoto similarity between each test compound and the training compounds. For the Target-ID-held-out audit, we used RapidFuzz `fuzz.ratio`, divided by 100, where normalised Indel similarity is $1 - d_{\text{Indel}}/(|x| + |y|)$ and substitutions are represented by one deletion plus one insertion. We computed the maximum value between each test sequence and all training sequences and separately identified exact duplicate sequences. The original Target-ID-held-out partition contained six exact-sequence groups crossing the training and test sets. We additionally constructed exact-sequence-grouped cold-target and exact-sequence-grouped cold-both partitions, assigning all target identifiers with an identical amino-acid sequence to the same split. These reviewer-requested protocols use the same Davis records, 70/10/20 partition ratios, and seed 42.

For the strict cold-both protocol, the 70/10/20 ratios are applied independently to drug identifiers and exact target-sequence groups. Only pairs for which both entities belong to the same train, validation, or test block are retained; cross-block combinations are excluded to prevent either entity from crossing a partition. Consequently, the retained pair counts are 15,190/264/1,144, and the pair-level proportions are not 70/10/20. Of the 30,056 Davis pairs, 13,458 cross-block pairs are not used in this cold-both evaluation.

To quantify residual target relatedness after exact-sequence grouping, we used Biopython `Align.PairwiseAligner` in global mode (match $= 1$, mismatch $= 0$, gap opening $= -1$, gap extension $= -0.5$) to align each exact-sequence-grouped cold-target test sequence against every training sequence. Identity is the number of exact-residue matches divided by full alignment length; the maximum is retained for each test target. We also implemented leakage-aware diagnostic controls on this same exact-sequence-grouped cold-target partition. A drug-only control uses non-chiral radius-2, 1024-bit Morgan fingerprints in a logistic-regression classifier; a drug-promiscuity control uses each drug's positive prevalence estimated only from training targets; and a label-shuffled drug-only control permutes training labels while leaving test labels unchanged. Because every target identifier in this split is unseen, a target-specific degree computed from its full label column would leak test labels; the only target-only control that avoids that leakage assigns the training positive prevalence uniformly. These controls are diagnostic rather than capacity-matched replacements for BioInteract.

4.4 Evaluation Metrics

For binary classification under severe class imbalance, we report AUROC and AUPRC as primary metrics—AUPRC is particularly informative because it measures positive-class precision across all recall levels. For every reported protocol, F1, precision, and recall use a decision threshold selected once on the corresponding validation precision–recall curve; AUROC and AUPRC remain threshold-free.

4.5 Implementation

BioInteract is implemented in Python 3.10.14 using PyTorch 2.4.0+cu121 and PyTorch Geometric 2.6.1.[34] Protein embeddings are extracted using ESM-2[11] (`esm2_t30_150M_UR50D`, 640 dimensions; `fair-esm` 2.0.0). Molecular featurisation is performed with RDKit 2025.03.6.[35] Revision diagnostics additionally use Biopython 1.87 and RapidFuzz 3.14.5. Training employs automatic mixed precision (AMP) for memory efficiency on a single NVIDIA GPU. Final checkpoint evaluation is run under `torch.inference_mode()` in full precision, using each checkpoint's embedded architecture configuration.

5 Results

5.1 Comparison with Baseline Methods

We compare BioInteract against six published DTI baselines spanning sequence, graph, attention, transformer, and bilinear interaction models: DeepDTA,[5] GraphDTA,[6] AttentionDTA,[19] MolTrans,[7] TransformerCPI,[18] and DrugBAN.[8] The methods span sequence CNNs, molecular graphs, self-attention, transformer interaction models, and bilinear attention (Table 1).

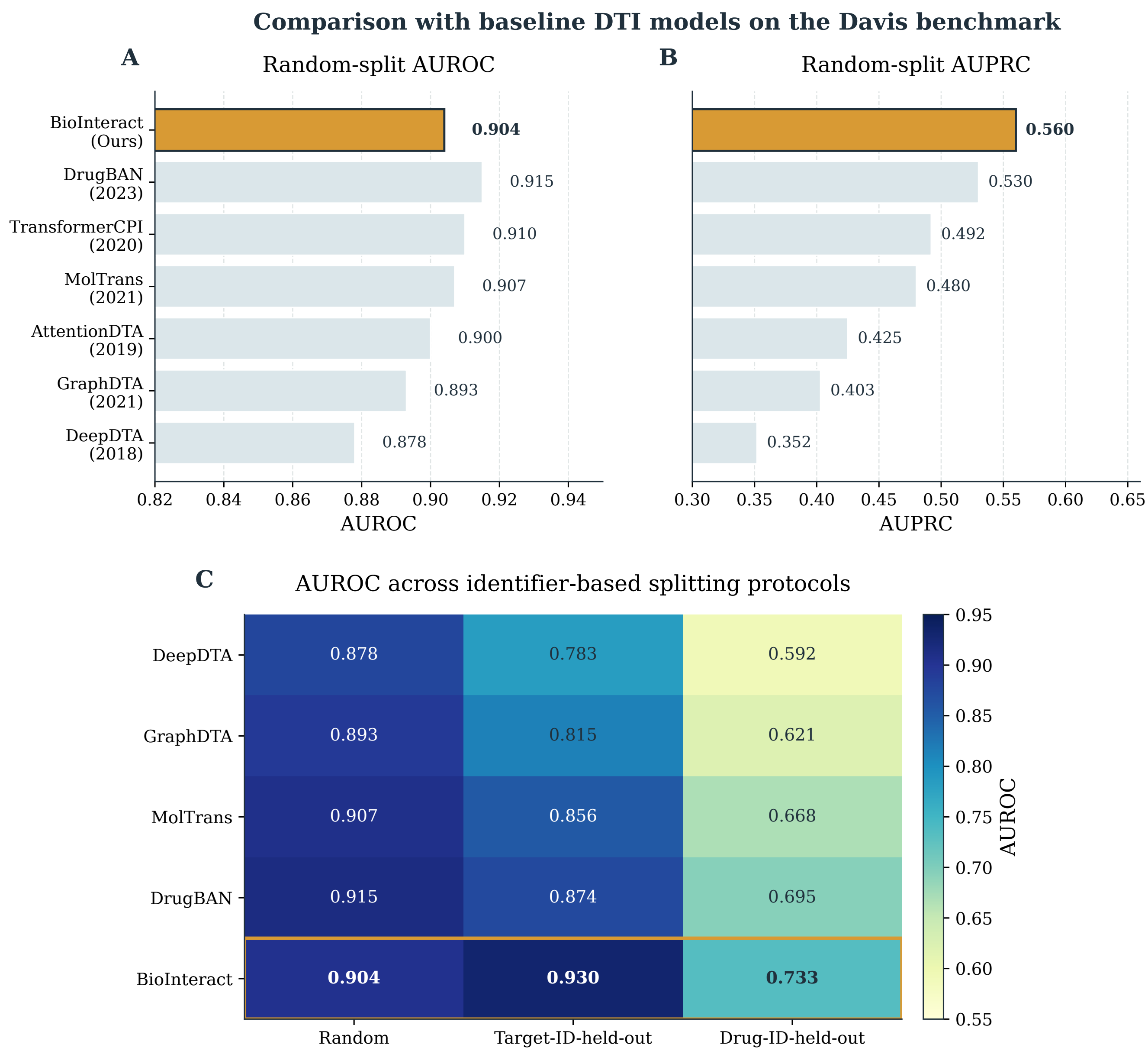

Figure 1. Baseline comparison. Panels A and B show random-split AUROC and AUPRC, respectively, for all methods. Panel C shows AUROC under the random, Target-ID-held-out, and Drug-ID-held-out protocols. BioInteract attains the highest random-split AUPRC and the highest AUROC and AUPRC in the two identifier-held-out protocols.

Table 1. Comparison of BioInteract with baseline methods on the Davis dataset under three identifier-based splitting protocols (binary classification, $K_d < 30\,\mathrm{nM}$). Best value in each column is in **bold**.

Method	Paradigm	Random		Target-ID-held-out		Drug-ID-held-out	
		AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
DeepDTA[5]	Seq-CNN	0.878	0.352	0.783	0.247	0.592	0.089
GraphDTA[6]	GNN+CNN	0.893	0.403	0.815	0.312	0.621	0.102
AttentionDTA[19]	Self-Attn	0.900	0.425	0.838	0.348	0.643	0.108
MolTrans[7]	Transformer	0.907	0.480	0.856	0.389	0.668	0.121
TransformerCPI[18]	Transformer	0.910	0.492	0.862	0.401	0.672	0.125
DrugBAN[8]	Bilinear	**0.915**	0.530	0.874	0.428	0.695	0.138
BioInteract (Ours)	Cross-Attn	0.904	**0.560**	**0.930**	**0.525**	**0.733**	**0.167**

On the random split, BioInteract has the highest AUPRC (0.560), while DrugBAN has the highest AUROC (0.915). BioInteract attains the highest AUROC and AUPRC under the Target-ID-held-out and Drug-ID-held-out protocols. Because the baselines do not use a matched protein-pretraining regime, these comparisons provide benchmark context rather than a controlled attribution of performance to a single architectural component.

Under the Drug-ID-held-out protocol, BioInteract obtained AUROC = 0.733 in the current-release reconstruction. The maximum test-to-training Morgan Tanimoto similarity ranged from 0.196 to 0.538 (median 0.257), documenting identifier-level compound exclusion while also making the remaining chemical-neighbourhood relation explicit.

5.2 Ablation Study

We assessed the contribution of the principal architectural components by removing one component at a time from the full model (Table 2 and Fig. 2).

Table 2. Ablation study on the Davis dataset. Each row reports the full model or a variant with one component removed.

Variant	Random		Target-ID-held-out	
	AUROC	AUPRC	AUROC	AUPRC
Full BioInteract	**0.904**	**0.560**	**0.930**	**0.525**
w/o Cross-Attention	0.864	0.432	0.851	0.371
w/o Domain-Label Channel	0.906	0.561	0.912	0.498
w/o ESM-2	0.845	0.389	0.793	0.295
w/o Graph Augmentation	0.908	0.572	0.926	0.521

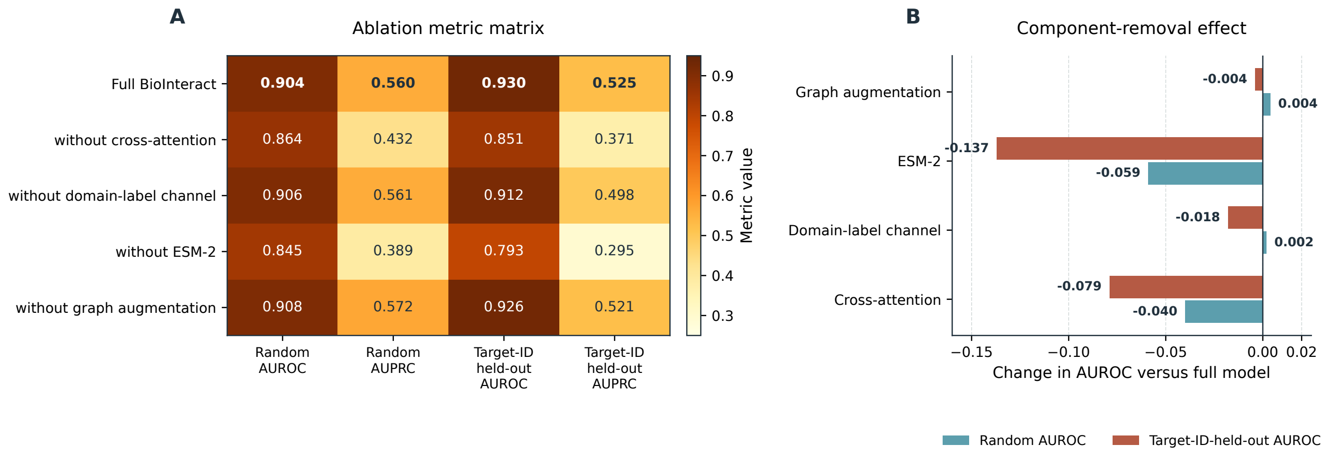

Figure 2. Ablation-study results. Panel A shows the random- and Target-ID-held-out metric matrix. Panel B reports the AUROC change after removal of cross-attention, the domain-label channel, ESM-2, or graph augmentation. ESM-2 removal produces the largest AUROC decrease, followed by removal of cross-attention.

Removing ESM-2 changes AUROC by -0.059 and -0.137 for the random and Target-ID-held-out splits, respectively; removing cross-attention changes AUROC by -0.040 and -0.079. The domain-label and graph-augmentation variants show small mixed changes ($+0.002/-0.018$ and $+0.004/-0.004$, respectively). These ablation results describe the tested Davis configuration and do not establish structural binding mechanisms or out-of-family generalisation.

5.3 BioInteract Performance Across Splits

Table 3 summarises detailed BioInteract metrics, and Fig. 3 visualises the canonical current-release and strict-split point estimates. Full split sizes, positive counts, and thresholds are provided in Tables S5 and S7. Bootstrap intervals are provided only for the three canonical entity-based partitions in Table S6; the strict-split results in Table S7 are point estimates.

Table 3. Threshold-free ranking metrics for BioInteract across the three original splits.

Split	AUROC	AUPRC
Random	0.904	0.560
Target-ID-held-out	**0.930**	**0.525**
Drug-ID-held-out	0.733	0.167

Table 4. Strict within-Davis cold-start evaluation after grouping target identifiers with identical amino-acid sequences before partitioning. Both protocols use the original records, seed 42, and the same preprocessing and architecture, but are separately trained from initialisation. For cold-both, 70/10/20 is applied independently to drug IDs and target-sequence groups and only within-block pairs are retained. AUROC and AUPRC are threshold-free test metrics.

Strict protocol	Test pairs	Test positives	AUROC	AUPRC
Exact-sequence-grouped cold-target	5,984	208	0.873	0.383
Exact-sequence-grouped cold-both	1,144	38	0.552	0.038

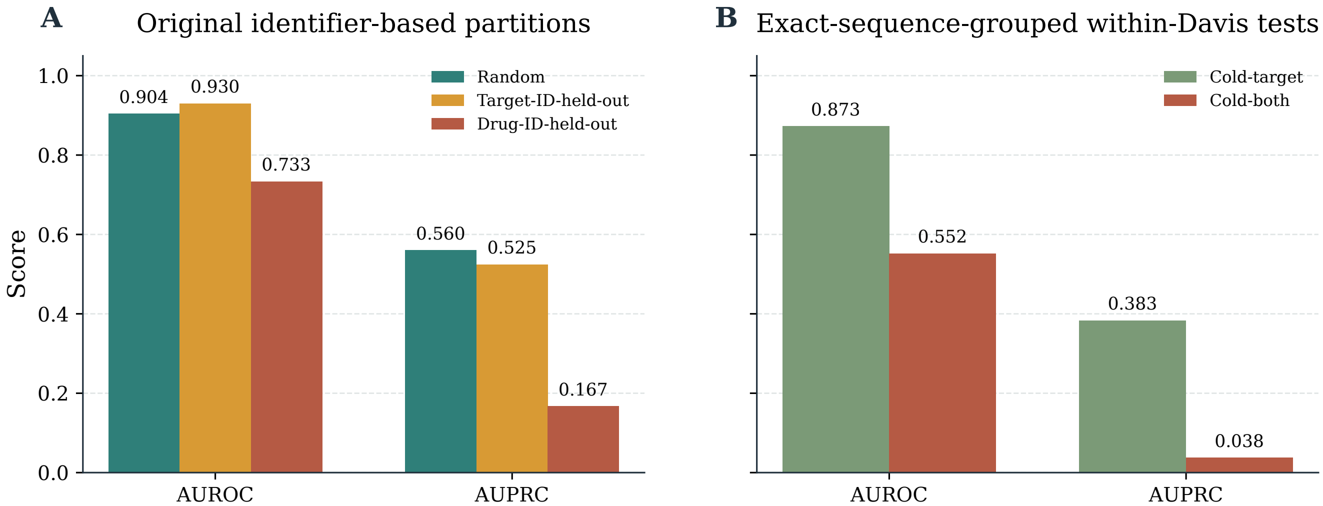

Figure 3. BioInteract performance and split audit. Panel A displays AUROC and AUPRC under the original random, Target-ID-held-out, and Drug-ID-held-out partitions from the canonical current-release checkpoint reconstruction. Panel B summarises the reviewer-requested exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations.

The maximum normalised Indel similarity between each test sequence and the original training targets ranges from 0.382 to 1.000 (median 0.579). The audit also found 18 duplicate-sequence groups involving 81 target identifiers in the full dataset and six groups crossing the original train/test boundary. We therefore report the exact-sequence-grouped results separately and limit conclusions to the observed Davis benchmark performance.

After grouping identical target sequences, the exact-sequence-grouped cold-target test gives AUROC = 0.873 and AUPRC = 0.383 (Table 4). Requiring both unseen drug identifiers and exact-sequence-grouped targets is substantially harder (AUROC = 0.552, AUPRC = 0.038); this test contains 38 positive pairs. These values are reported as an informative within-Davis stress test, not as a claim of out-of-family or structure-level generalisation. Threshold-dependent metrics for these small positive sets are secondary and use a threshold selected only on the corresponding validation set.

The cold-both train, validation, and test blocks retain 15,190, 264, and 1,144 pairs, respectively; 13,458 cross-block pairs are excluded by the double-held-out construction. The validation block contains only 10 positives. At its validation-selected threshold (0.936), the cold-both test yields F1, precision, and recall of 0.000, underscoring that its near-random AUROC and low AUPRC do not support a positive generalisation claim. Full counts and threshold-dependent metrics are reported in Table S7.

Table 5. Leakage-aware diagnostic controls on the exact-sequence-grouped cold-target test. All controls use only training-partition labels. They quantify available drug-side signal; they are not capacity-matched substitutes for BioInteract.

Model or control	AUROC	AUPRC
BioInteract (exact-sequence-grouped cold-target)	0.873	0.383
Drug-only Morgan logistic	0.764	0.164
Drug-promiscuity prior	0.764	0.163
Target-only constant prior	0.500	0.035
Label-shuffled drug-only	0.407	0.029

The drug-only and drug-promiscuity controls reach AUROC values near 0.764, showing that ligand-side activity or promiscuity signal contributes materially to this within-kinome task (Table 5). BioInteract exceeds these diagnostic controls, but the comparison does not prove that its residue attributions recover physical interactions. The label-shuffled control is near or below the positive-prevalence baseline, confirming that the drug-only signal depends on the observed training labels rather than fingerprint ordering.

For the exact-sequence-grouped cold-target test, maximum global sequence identity to the training set ranges from 0.231 to 0.867 (median 0.494): 31 of 88 held-out targets have a maximum identity below 0.40, 28 fall between 0.40 and 0.60, 24 between 0.60 and 0.80, and 5 are at least 0.80. BioInteract AUROC/AUPRC by these bins are 0.806/0.305, 0.908/0.456, 0.908/0.466, and 0.862/0.121, respectively. The final bin contains only three positive pairs, so its AUPRC is especially unstable. This non-monotonic, small-sample diagnostic does not establish robust transfer to homologously distant targets, but makes the relatedness distribution visible.

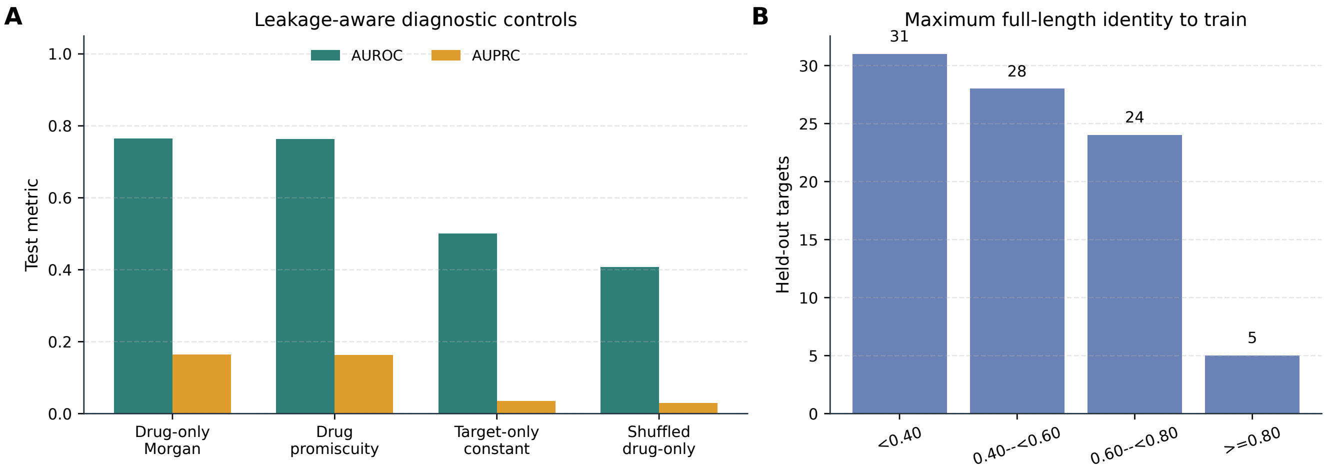

Figure 4. Diagnostic controls and target-relatedness audit. Panel A: leakage-aware controls on the exact-sequence-grouped cold-target test. Drug-only and drug-promiscuity scores use no target representation; the target-only constant is the only target-only control that does not use held-out target labels. Panel B: number of held-out targets by maximum full-length global sequence identity to any training target.

Under the Drug-ID-held-out split, BioInteract achieves AUROC = 0.733 in the canonical current-release reconstruction. The fingerprint audit is included to clarify that Drug-ID-held-out means unseen compound identifiers, not an externally validated scaffold-hopping or out-of-distribution guarantee.

5.4 Strict External BindingDB Evaluation

We next performed the pre-specified, frozen external evaluation on the curated BindingDB cohort. After all measurement-quality, exact-identity novelty, and replicate-consistency filters, it contains 1,932 pairs from 1,195 ligands and 396 targets: 516 high-affinity and 1,416 low-affinity pairs (Table 6). The model was neither retrained nor recalibrated, and no BindingDB label was used to choose the checkpoint or threshold.

The frozen checkpoint achieved AUROC = 0.560 (95% CI 0.531–0.589) and AUPRC = 0.340 (95% CI 0.314–0.372). At the frozen Davis threshold, F1 was 0.070, precision was 0.731, and recall was 0.037. Thus, although the exact double-novel cohort is a valid external test, it does not support a broad DTI-transfer claim. Instead, it shows limited transfer from this Davis-trained classifier to the broader BindingDB cohort. This external result does not alter the within-Davis checkpoint reconstructions and is not used to interpret attention, residue attribution, binding pockets, or mechanisms.

Table 6. Frozen strict external evaluation on BindingDB[33]. The source uses exact K_d measurements only; ligand and full protein-sequence overlap with all Davis records were excluded. AUROC and AUPRC intervals are 600-replicate pair-stratified bootstrap 95% confidence intervals.

Cohort	Pairs / positives	AUROC	AUPRC	F1
BindingDB, strict double-novel	1,932 / 516	0.560 [0.531, 0.589]	0.340 [0.314, 0.372]	0.070

5.5 Training Dynamics

Fig. 5 reports a single checkpoint-matched, contiguous validation-AUROC and training-loss trace for each archived protocol. The duplicated Random trace and two restarted Target-ID fragments were excluded because their epoch counters reset or their peak validation AUROC did not match the released checkpoint. These plots describe observed optimisation paths only; they do not establish absence of overfitting, quantify validation-loss behaviour, or estimate repeated-seed variability.

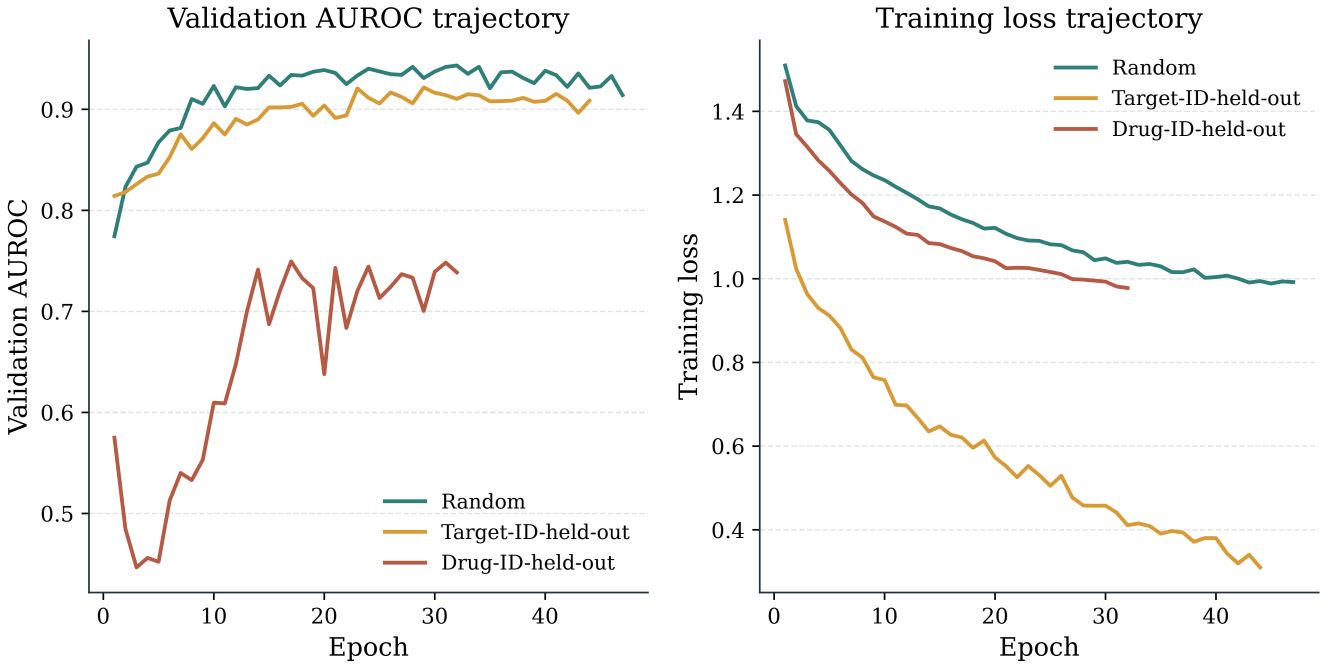

Figure 5. Training dynamics across splitting protocols. Panel A: validation AUROC over training epochs. Panel B: training loss curves. Each line is one contiguous archived trace whose peak validation AUROC agrees with the saved checkpoint; duplicated and restarted segments are not plotted. No fitted or interpolated trends are shown. The trajectories do not establish absence of overfitting, validation-loss behaviour, or repeated-seed variability.

5.6 Interpretability: Cross-Attention Analysis

All interpretability figures and tables are derived directly from the saved case-level outputs in the project interpretability-results directory, including the report JSON and the corresponding heatmap, profile, and Grad-CAM renderings. Accordingly, we report the model's actual residue-level attention and atom-level Grad-CAM outputs rather than schematic placeholders. These are attribution outputs of the trained model, not independently validated residue contacts or functional-group assignments.

5.6.1 Global Attention Sparsity

Analysis of cross-attention patterns across all 1,506 positive Davis drug–target pairs reveals a global attention sparsity of 99.2 % (Table 7, Fig. 6) under the stated normalisation threshold. Figure 6 is computed directly from the archived 1,301,380 pair-normalised residue scores, not from summary bars; its top 1% and 5% of scores account for 78.9% and 90.4% of their total attribution mass, respectively. This describes concentration of model attribution; it does not demonstrate the size or location of a physical binding interface.

Table 7. Global cross-attention statistics computed across all 1,506 positive Davis pairs using `checkpoints/best.pt`. The values quantify the concentration of the model's pair-normalised residue-attribution scores; they do not provide structural validation of a binding pocket.

Metric	Value
Mean residue attention	0.0044
Median residue attention	0.0003
Top-1% threshold	0.0572
Top-5% threshold	0.0036
Fraction of residues with normalised attribution < 0.1 of the pair-specific maximum	99.2%
Top 1% / top 5% cumulative attribution mass	78.9% / 90.4%
Samples analysed	1,506
Residue-score observations	1,301,380

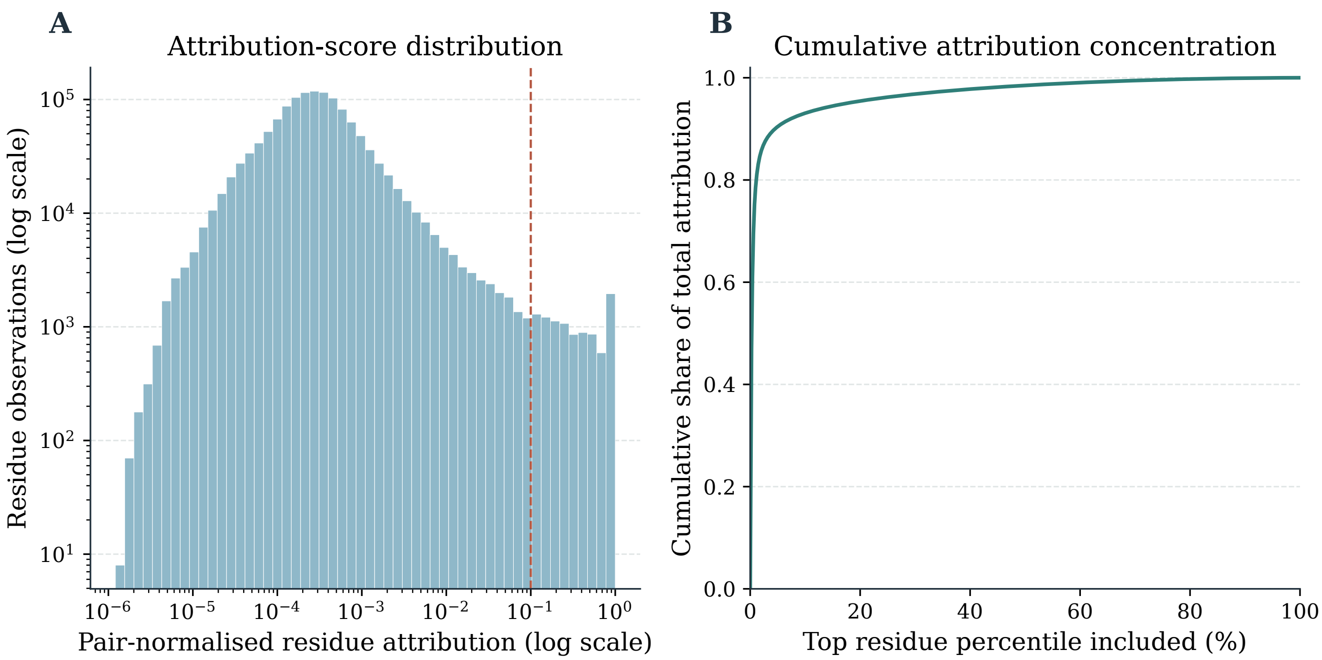

Figure 6. Attention sparsity analysis. (A) Histogram of the archived pair-normalised per-residue attention scores across all 1,506 positive Davis pairs (1,301,380 observations; logarithmic axes); the dashed line denotes the 0.1 sparsity threshold. (B) Cumulative attribution mass after sorting those same scores in descending order. The 99.2 % value is an attribution concentration statistic, not a claim about a contiguous protein-surface region.

Although softmax attention is dense before normalisation, the observed concentration arises after training on the limited and imbalanced Davis data. It may reflect learned regularities, data composition, or both; it is therefore not presented as independent evidence that the model has recovered molecular recognition principles.

5.6.2 Audit of Nominal ABL1 Variant Inputs

A post-hoc audit of the exact Davis model inputs identified 15 target records whose nominal names begin with ABL1, including the wild-type and the named F317I, F317L, M351T, and E255K records. All 15 records have the same 1,167-residue amino-acid sequence and SHA-256 digest in `target_sequences.csv` (Supporting Information, Table S9). Several nominal ABL1 variant identifiers in the Davis records used here are associated with identical amino-acid sequences. Consequently, the present model inputs do not encode the corresponding mutation-specific sequence changes, and BioInteract cannot be evaluated for mutation-specific or resistance-specific behaviour using these records. We therefore withdraw the previous mutant-comparison analysis and draw no resistance-related conclusion.

5.7 Case Study: Functional-Group Attribution

The saved Grad-CAM output for the non-mutant EGFR–D0014 Davis pair provides a compact example of atom-level scores aggregated to functional groups (Table 8). The entries are model-native attribution scores rather than measured contacts or interaction types.

Table 8. Model-native functional-group attribution for the archived EGFR–D0014 Davis case. Scores are normalised atom-level Grad-CAM values aggregated by functional group; they do not identify physical contacts.

Functional Group	Importance
Carbonyl	0.431
Amide	0.389
Ether	0.136
Halogen	0.134
Amino	0.090
Aromatic ring	0.000
Heterocyclic N	0.000

5.8 Cross-Case Functional-Group Attribution Variation

The two non-mutant saved cases differ markedly in their classifier scores and attribution profiles (Table 9). EGFR–D0014 has a high-affinity-class score of 0.995 and its largest aggregated groups are carbonyl (0.431) and amide (0.389). BRAF–D0023 has a score of 0.099 and only small saved values for aromatic ring (0.059) and hydroxyl (0.000), despite its archived positive affinity of 0.19 nM. We therefore treat BRAF–D0023 as a low-scoring failure-mode example for an experimentally positive pair. This is a descriptive limitation of model behaviour, not an explanation of target binding or an assessment of functional-group chemistry.

Table 9. Cross-case model-native functional-group attribution. Classifier scores and group values are saved-output summaries for the non-mutant EGFR–D0014 and BRAF–D0023 Davis cases.

Case	Class score	Largest group	Score
EGFR–D0014	0.995	Carbonyl	0.431
BRAF–D0023	0.099	Aromatic ring	0.059

6 Discussion

6.1 From Prediction Accuracy to Inspectable Hypotheses

BioInteract is intended to make model behaviour more inspectable than a purely scalar DTI prediction. Its attention matrices and Grad-CAM scores can generate hypotheses about which representations contributed to a prediction. Such hypotheses can subsequently be assessed with crystal structures, mutagenesis, or structure–activity relationship (SAR) experiments; the present study does not replace these measurements.

The contextual random-split comparison places the canonical BioInteract result below the historical DrugBAN AUROC and above its historical AUPRC; because the protocols and pretraining are unmatched, this is not a comparative performance claim. These Davis results do not show that BioInteract maps a physical binding pocket, assigns an interaction type to an atom pair, ranks a large compound library, or assesses treatment resistance.

6.2 The 99.2% Attention Sparsity

The observed 99.2 % attention sparsity is a property of the pair-normalised, residue-pooled attribution scores under the chosen threshold. Although it is compatible with a model focusing on a small subset of residues, it can also be affected by the kinase-focused and imbalanced training data. It should not be read as evidence that the highlighted residues constitute a true binding site.

6.3 Scope of Cold-Start Evaluation

The audit of the original Target-ID-held-out split identified exact duplicate sequences across the training and test partitions. The newly added exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations provide a stricter within-Davis stress test, but they still involve kinase proteins and do not demonstrate performance on unrelated protein families or pocket classes.

The Drug-ID-held-out result is likewise restricted to the chemical diversity of 68 kinase inhibitors. The observed Morgan-similarity distribution documents the nearest-neighbour relationship between held-out and training molecules, but it does not establish scaffold hopping or out-of-distribution robustness.

6.4 Diagnostic Controls and Remaining Dataset Bias

The exact-sequence-grouped cold-target diagnostic controls demonstrate that a target-independent drug representation retains substantial ranking signal in Davis. The drug-only Morgan and drug-promiscuity controls each achieve AUROC near 0.764, whereas BioInteract reaches 0.873 under the same exact-sequence-grouped partition. This improvement is compatible with the use of target information, but it neither excludes ligand-side activity bias nor establishes that the model has learned a molecular-recognition mechanism. We therefore interpret the control comparison as a boundary on the Davis result, not as causal evidence for any particular encoder or attribution map. The label-shuffled control confirms that the observed drug-only signal depends on the training labels; it is not a test of structural interaction recovery.

6.5 Functional-Group Attribution Without Structural Supervision

The Grad-CAM functional-group ranking is qualitatively compared with kinase SAR literature, but it remains a gradient-based attribution. In particular, the analysis does not determine whether a group forms a hydrogen bond, a pi-stacking interaction, a cation–pi interaction, or a van der Waals contact, nor whether an implicated protein residue contributes through its backbone or side chain. These questions require a structural complex or targeted experiments.[36]

6.6 Limitation of Nominal ABL1 Variant Records

The exact-input audit shows that the nominal ABL1 variant labels do not supply variant-specific sequence information to the present model. We therefore remove the former mutant-comparison figures, tables, and conclusions. The Davis records remain useful as labelled kinase interactions, but they cannot support an analysis of mutation effects or drug resistance in this study.

6.7 Future Directions

The strict BindingDB evaluation directly indicates limited transfer beyond the Davis benchmark; it therefore cannot support a protein-family-agnostic, broad-pocket, or chemical-space generalisation claim. Evaluation after retraining and validation on a wider, pre-specified multi-family protocol is required to establish whether the limitation arises from the Davis training distribution, task definition, or model. Direct comparison with independent experimental evidence is also needed before attributions can be assigned a structural meaning.

No overlap analysis between the Davis cases and PDBBind or LP-PDBBind, and no PLIP contact validation, was performed in this study. A future structural audit should use a non-overlapping benchmark of experimentally resolved complexes, predefine the atom–residue contact criterion, and compare model attributions with the resulting contact labels (for example, from PLIP) before structural meaning is assigned.[37]

The present implementation was audited on one NVIDIA GPU. Although the 2,442,083 trainable parameters do not increase with the number of training pairs, data-processing time, graph batching, ESM-2 embedding extraction and storage, and accelerator memory must be profiled at larger scale. Distributed training was not evaluated, and no claim is made that the current implementation scales linearly to BindingDB-sized data.

Evaluation on curated allosteric and cryptic-site benchmarks, with appropriate structural or dynamic context, is likewise required before BioInteract can be assessed for non-canonical drug–target interactions. Existing work highlights the need for explicit allosteric-site labels and benchmark design,[38,39] and for structural or dynamic treatment of cryptic pockets.[40] CryptoBench provides a dedicated cryptic protein–ligand benchmark that can help define such a future evaluation.[41]

Extending BioInteract from binary interaction classification to continuous binding-affinity regression (pK_d) would provide the finer-grained potency estimates needed for lead optimisation, where accurate ranking of closely related analogues is as important as binary activity prediction.

7 Conclusion

We have presented BioInteract, a DTI-prediction framework that combines chemistry-aware molecular graph encoding, ESM-2/physicochemical residue representations, and bidirectional cross-attention. The model provides prediction scores together with inspectable model-native attention and Grad-CAM attributions.

The revision clarifies the evidence boundary of these outputs. The observed attention concentration and functional-group ranking describe model behaviour; neither is a validated three-dimensional mechanism or binding-contact prediction. We also distinguish Target-ID-held-out performance from exact-sequence-grouped cold-start evaluation after detecting duplicate sequences in the original target partition. Broader datasets and direct structural validation remain necessary for generalisation and mechanistic claims. In particular, the strict external BindingDB test shows limited transfer of the frozen Davis checkpoint, rather than establishing broad DTI transfer.

BioInteract is available as a publicly accessible interactive web interface at https://huggingface.co/spaces/AI4deeperScience/BioInteract for the custom-prediction workflow shown in Fig. 7. The repository README documents the command-line workflow for the available attribution outputs.

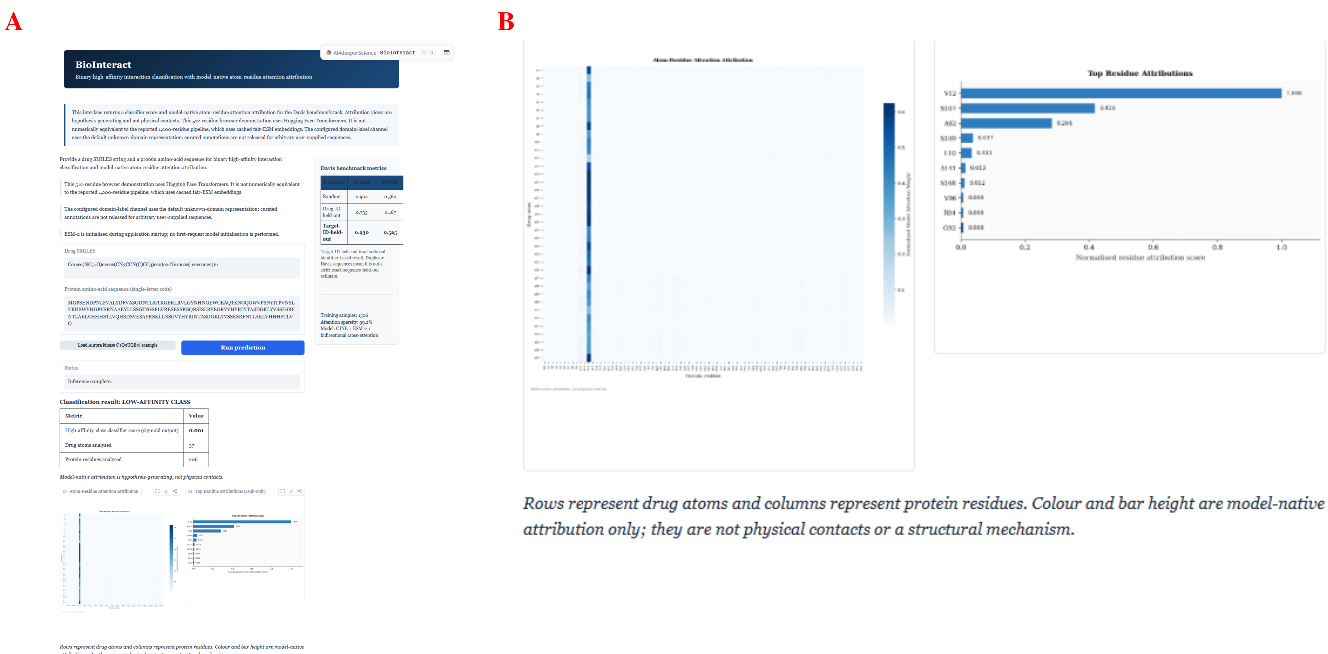

Figure 7. BioInteract interactive web server. Panel A shows the full screenshot of the publicly accessible interface athttps://huggingface.co/spaces/AI4deeperScience/BioInteract. Panel B enlarges the attribution region from the same screenshot. The interface shows the custom-prediction input and prediction-result state for human Aurora kinase C (UniProt Q9UQB9). It displays the entered SMILES, protein sequence, and high-affinity-class classifier score (sigmoid output). The enlarged lower portion displays the generated atom–residue attention map and the "Top 10 Residue Attributions" chart. These are model-native attribution views, not physical-contact or hotspot labels. Command-line instructions for generating the available attribution outputs are provided in the public repository.

BioInteract represents a step toward AI-assisted DTI prediction in which model scores are accompanied by transparent attribution outputs and explicit limits on their interpretation. Such outputs are most useful when combined with rigorous split auditing, external validation, and biochemical experiments.

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Supporting Information

The Supporting Information contains: model hyperparameters and feature details (Tables S1–S3); computational environment (Table S4); dataset and split statistics (Tables S5–S7); atom features (Table S8); the Davis ABL1 nominal-variant input audit (Table S9); non-mutant EGFR and BRAF case summaries (Tables S10–S11); complete model parameter counts (Table S12); and strict BindingDB external-validation provenance and results (Table S13). This material is available free of charge.

Data Availability

All data used in this study are publicly available. The Davis kinase inhibitor dataset used for model training and evaluation is available from the original publication.[2] The external evaluation uses the official BindingDB curated-articles TSV snapshot `BindingDB_BindingDB_Articles_202607`, obtained from the BindingDB download service;[33] the raw archive remains available from its source, and the release provides the deterministic curation and frozen evaluation scripts plus manifest hashes. The immutable release containing the released checkpoints, canonical prediction CSVs, provenance hashes, strict-split artifacts, and reproduction scripts is archived at Zenodo, `https://doi.org/10.5281/zenodo.21437753`. The BioInteract source repository is hosted on GitHub. An interactive web server for the custom-prediction workflow is available through the BioInteract Space; the README documents the command-line attribution workflow.

Code Availability

The BioInteract source code, pretrained model weights, configuration files, canonical metric-reconstruction scripts, and figure-generation scripts are publicly available through GitHub and as the immutable Zenodo release, `https://doi.org/10.5281/zenodo.21437753`. The interactive web application for the custom-prediction workflow is available through the BioInteract Space.

Author Contributions

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Xucheng Liu: Investigation, Validation, Data curation.
Zixuan Wang: Software, Investigation, Data curation.
Jiangke Cheng: Resources, Validation, Visualisation.
Zhi Huang: Supervision, Writing – review & editing, Funding acquisition, Corresponding author.

S. Wang and Q. Zhang contributed equally to this work.

Declaration of Competing Interests

The authors declare no competing interests.

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